Faculty of Science

Active Learning Guided High-Throughput Design of Multi-Metal Oxides for Acidic Oxygen Evolution Reaction

A *Research Proposal* submitted by

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Content

1. Background

The oxygen evolution reaction (OER) plays a vital role in energy storage and conversion technologies such as solar/electricity-driven water splitting, rechargeable metal–air batteries, and regenerative fuel cells. However, the OER in these electrochemical systems encounters sluggish kinetic limitations and lead to high overpotentials due to a multistep proton-coupled electron transfer mechanism.

Metal oxide (MO) electrocatalysts have received considerable interest due to their **compositional/structural diversity.** During the past decade, metal oxides in three common structural forms, i.e., simple BO_x transition metal oxides, B_3O_4 spinel oxides and ABO_3 perovskite oxides have been widely studied and exhibit great catalytic activity[1]. Recently, some complex oxides with special crystal structures or multiple metal coordination environments have been reported as promising OER catalysts[2], including triple/quadruple perovskites, brownmillerites, pyrochlores, etc. More strikingly, developing 2D[3], layered[4], and amorphous[5] structured MOs have proven to be an effective method for high-performance OER catalysts. In addition to structural innovation, introducing multi-principal element compositions holds the potential for synergistic catalytic effects. Recent efforts[6–8] aim to design multi-metallic oxide OER catalysts that transcend conventional trade-offs between activity and stability.

Another key advantage of MOs lies in their **flexible tunability.** Various material engineering strategies ranging from atomic to mesoscale level have been applied to enhance OER performance. Among them, the modulation of point defects, such as heteroatom doping and oxygen-vacancy control, has been extensively explored in various MO electrocatalysts to tailor their catalytic properties. Specifically, defects serve to: (i) optimize the adsorption/desorption dynamics of intermediates to accelerate kinetics, (ii) direct surface reconstruction to expose more active sites, and (iii) switch the reaction pathway from the adsorbate evolution mechanism (AEM) to the lattice oxygen mechanism (LOM). In addition, point defects contribute to long-term stability and functionality in harsh catalytic environments.

Ru/Ir-based oxides are universally identified as the benchmark OER electrocatalysts[9], but their high cost and scarcity limit their widespread applications. To date, the average price of iridium is US \$58,000 lb^{-1} over the five-year period from January 2019 to January 2024, which is significantly higher than that of cobalt (US\$19 lb^{-1}), iron (US\$0.26 lb^{-1}) and manganese (US\$1.5 lb^{-1})[10]. Therefore, extensive research has been conducted to develop non-noble alternatives[11,12]. Unfortunately, earth-abundant metal-based oxide catalysts perform well in the alkaline OER (pH ∼14) but often suffer stability issues in acidic environments (pH ∼0).

The exploration of complex multi-metallic systems for acidic OER remains in its early stages, largely constrained by their inherent compositional and structural complexity. Prior efforts via high-throughput (HT) computations demonstrate the viability of narrow down possible candidates before synthesizing them in a lab[13]. However, exhaustive, brute-force searches become intractable as the number of oxide polymorphs and constituent elements increases exponentially. The integration of ML/AI algorithms into HT workflows holds considerable promise for more effective catalyst design, yet realizing their full potential requires greater transparency, scalability, and flexibility[14].

2. Objectives

We intend to achieve the following goals: (i) Search for acid OER catalysts based on existing datasets that outperform the currently available materials by evaluating thermodynamic, electrochemical, and kinetic properties. (ii) Develop curated computational datasets, a high-throughput pipeline, an active learning framework, and trained ML models for acid OER catalyst discovery. (iii) Establish a quantitative mapping of structure-property-performance relationships for a diverse set of oxide compositions and structures.

3. Proposed Work

By harnessing the collective power of density functional theory (DFT) and multiple machine learning techniques, these research efforts will be organized into four key stages: (i) computational methodologies development, (ii) candidate library generation, (iii) active learning aided high-throughput screening, followed by (iv) uncovering structure-electronic-activity relationships.

2.1 Computational Methodologies Development

This section outlines the proposed DFT calculation methodologies and settings for MO electrocatalysts under acid OER conditions.

2.1.1 Bulk Acid Stability

Thermodynamically, the energy above the convex hull (E_{hull}) represents the energy of decomposition of a material into the set of the most stable phases at the same chemical composition. This thermodynamic phase stability can be used as a metric to estimate the synthesizability of the candidate material. When the E_{hull} is higher, the material is more unstable and synthesis will be challenging.

Electrochemically, the Pourbaix diagram, also called potential-pH diagram, is a representation of aqueous phase electrochemical equilibrium. It shows water-stable phases as a function of pH and potential, where the potential is defined with respect to the standard hydrogen electrode (SHE). The diagram will be constructed by considering the thermodynamic equilibrium among all possible species in aqueous solutions including the elements, compounds, aqueous ions, and liquid water. We intend to adopt the scheme developed by Singh et al.[15] to determine free energies, which has been implemented in Python Materials Genomics (pymatgen)[16] package. The harsh OER conditions are defined as pH = 0 and U = 1.23 V_{RHE} .

2.1.2 Surface Energy

Surface energy calculations will be performed on various facets for slabs of increasing thickness. Then the bulk energy will be extracted by fitting the total energy of the slabs against the number of layers as explained in Fiorentini et. al.[17] The aim is to avoid common issues of surface energy divergence associated with using a separate bulk energy calculation.

2.1.3 Adsorption Energy and Overpotential

Building upon the computational hydrogen electrode model[18], the OER overpotential can be derived from Gibbs free energies of reaction intermediates (ΔGO*, ΔGOH*, ΔGOOH*):

$$\eta_{\mathrm{OER}}(\mathrm{V}) = \max[\Delta G_{\mathrm{OH}^*}, \Delta G_{\mathrm{O}^*} - \Delta G_{\mathrm{OH}^*}, \Delta G_{\mathrm{OOH}^*} - \Delta G_{\mathrm{O}^*}; 4.92 - \Delta G_{\mathrm{OOH}^*}]/\mathrm{e} - 1.23\ \mathrm{V}$$

To calculate the adsorption free energies of the reaction intermediates, DFT-calculated electronic energies should be converted into free energies by adding Harmonic oscillator approximation based free energy corrections (zero-point energies, enthalpic and entropic contributions), as shown in Table 1.

Table 1. Gibbs free energy correction values for reference gas phase species and adsorbates. The differences between zero-point energy (△ZPE), and entropy (ΔS) are calculated using vibrational frequencies. The unit used is eV

Adsorbates	ZPE	$\int C_p dT$	-TS	Molecules	ZPE	$\int C_p dT$	-TS
O^*	0.08	0.03	-0.06	H_2	0.28	0.09	-0.41
OH^*	0.37	0.04	-0.07	H_2O	0.57	0.10	-0.67
OOH^*	0.41	0.05	-0.09				

Gibbs free energies are then calculated by adding the correction values to the calculated DFT electronic adsorption energies:

$$\Delta G_{\text{adsorption}} = \Delta E_{\text{adsorption}} + \Delta G_{\text{correction}}$$

Additionally, all calculations will be performed with spin polarization to account for the spin states in metals. Despite the magnetic polymorphism observed in some oxide materials, considering only one polymorph for each slab is sufficient, which can be initialized with either ferromagnetic or nonmagnetic configurations, depending on the magnetic moments of each metal. To address the self-interaction errors of DFT for strongly correlated systems, we need to apply the Hubbard U correction scheme. Xu et al. reported that the corrections led to more consistent with experimental observations[19].

2.2 Candidate Library Generation

We plan to construct our search space by mining data from previous publications, including comprehensive databases (e.g., the Materials Project[20] and the Open Quantum Materials Database[21]) and catalyst-specific high-throughput data[22–25]. Past research has revealed that most of the acid-stable materials are ternary oxides, and the rest are quaternary oxides[26]. Therefore, single metal oxides will be excluded from our candidate pool. In addition to bulk properties, surface should be labeled with facet, termination, coverage, and adsorption site.

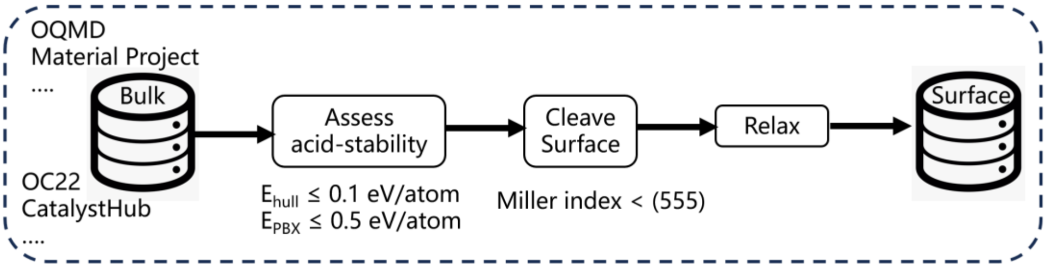

Fig.1 Automation pipeline for generating the surface library

2.2.1 Bulk Selection

The first criterion assesses the thermodynamic stability of the candidate material via E_{hull}. Simliar to Pourbaix stability, Materials with a calculated $E_{hull} \leq 0.1$ eV/atom have been shown to exhibit synthesis in experiment. The second criterion describes the Pourbaix stability. We will quantify the Pourbaix stability of the bulk using the Pourbaix decomposition energy (E_{PBX}). $E_{PBX} \leq 0.5$ eV/atom was chosen as the stability threshold. According to previous studies, applying these two criteria greatly reduces the search space. A computational analysis performed by Wang et al. indicated only 68 bimetallic oxides were stable from a pool of 47,814 ($E_{PBX} \leq 0.5$ eV, pH=0 and U=1.2-2.0 V)[26]. Back et al. reported 14 stable bulk candidates out of 2600 bimetallic combinations[24].

2.2.2 Slab Generation

There are multiple exposed facets with different O terminations, and these facets have multiple unique adsorption sites. Here we will outline the procedure used to generate slabs from stable bulk phases. The procedure was as follows: the slabs (isolated face) will be construct by enumerating low-index facets up to a

Miller index of (555), each containing an atomic and vacuum layer. Each of these slabs will be evaluated under three coverage limits: bare, O-covered and OH-covered. We assume the adsorbate can bind to two types of sites: the surface oxygen and the under-coordinated surface metal. Note that we neglect surface segregation and reconstruction.

2.3 Active Learning Accelerated High-Throughput Screening

Active learning in conjunction with Bayesian Optimization (BO) will be employed and synergistically coupled with high-throughput computational feedback. The BO approach is designed to be (1) responsive in improving itself by learning from small batches of newly acquired DFT data, and (2) aware of limitations in its surrogate model by incorporating uncertainty estimates into the acquisition criteria.

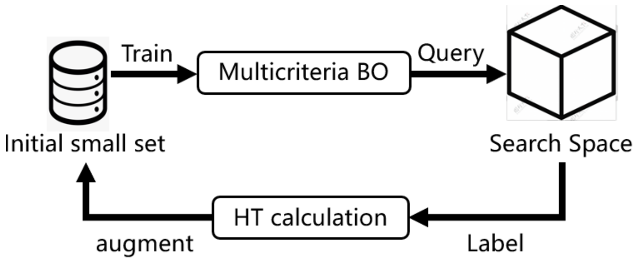

Fig 2. Schematic workflow of active learning loop

As mentioned earlier, high-performing electrocatalysts must combine high activity (low overpotential) and stability (low surface energy), which represents multicriteria optimization task. Existing computational catalyst screening approaches are multi-step processes, i.e., narrowing down the candidate's step by step considering multiple criteria including stability, activity, price, and toxicity, in a sequential manner. We seek out to develop AL strategy merging the screening steps for simultaneous multicriteria evaluation. Specifically, we will explore two advanced BO methods: constrained BO and multi-objective BO.

2.3.1 Constrained BO

The multicriteria screening can be treated as a constrained BO problem, involving optimizing the objective (finding a material targeting high activity) while not violating the constraints (e.g., stability should exceed a threshold). Therefore, the new sample will be selected with the highest constrained expected improvement for each iteration. Within the constrained BO framework, chen et al. proposed uncertainty-aware atomistic machine learning model, UPNet[27], which we intend to leverage for its automated representation learning and interpretable feature extraction in our screening task.

2.3.2 Multi-objective BO

The goal of multi-objective BO (MOBO) is to learn the Pareto front: the set of optimal trade-offs, where an improvement in one objective means deteriorating another objective. This aligns well with the observations in metal oxides electrocatalysis, where activity and stability tend to be inversely correlated. The MOBO approach has already proven successful in building both computational and experimental HT workflow[8,28].

2.3.3 Open-source Packages

Effective acquisition functions and surrogate models are readily supported in open-source libraries[29].

Table 2 Open-source BO packages

Library	Surrogate model	Acquisition model	Constraint	Multi-objective
BoTorch	GP	(L)PI, (L)EI, KG, PM, UCB, (M/P/J) ES,	✓	✓
Dragonfly	GP	SR, EHVI, MSLA, PV, EUBO	✓	-
COMBO	GP	EI, TS, UCB, PI	✓	✓
PHYSBO	GP	PI, EI, TS	-	-
BOSS	GP	PI, EI, TS, HPVI, EHVI	✓	✓
SMAC3	GP, RF	EI, ELCB	✓	✓

2.4 Machine Learning Structure-Property-Performance Relationships

We are also highly interested in exploring the site-specific responses of acid catalytic performance as a function of electronic structure parameters. For this purpose, we can create more active sites by decorating optimal surface candidates with point defects, i.e. oxygen vacancies and metal substitutions to build training dataset. Common transition-metal elements including Mn, Fe, Co, Ni, Cu, and Zn will be used as dopants. As a result, the adsorbate can bind to three types of sites: the surface oxygen, the under-coordinated surface metal, or an oxygen vacancy.

To describe site-dependent properties, we need to move beyond bulk-level descriptors towards a more realistic representation of catalytic sites. We will consider multiple DFT-computed per-site descriptors based on local electronic structure: magnetic moments, Bader charges, atomic vibration frequencies, metal d-band centers, and oxygen 2p-band. By utilizing these descriptors, ML surrogate models will be implemented to both predict per-site properties directly from structure and leverage these properties to predict OER activity. The ML surrogate model can be develop based on graph representations, which naturally satisfy permutation invariance. For reference, CGCNN[30] and SchNet[31] represent the two most prominent graph neural network architectures tailored for material science applications.

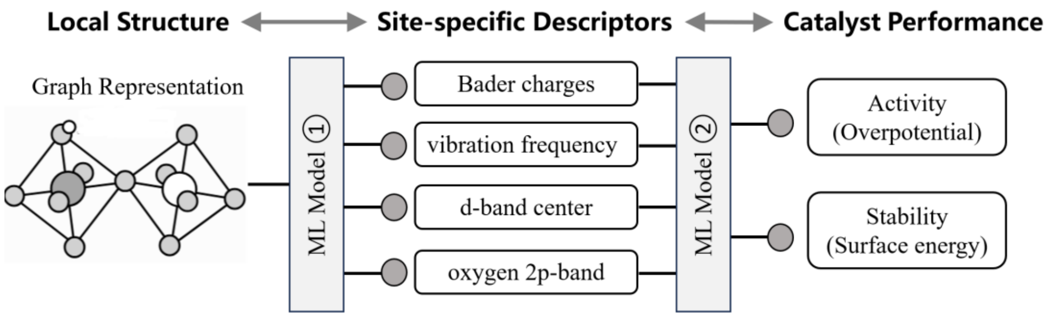

Fig 3. Machine learning mapping between local environment, per-site properties and catalyst performance

In addition to achieving excellent prediction accuracy, we also aim for ML surrogate models to provide insights into structure-electronic-activity/stability relationships and atom-level design. To this end, Shapley additive explanations (SHAP) analysis[32] will be carried out to quantify the contribution of features and samples to prediction outcomes across all possible combinations. We will extract chemical insights using following approaches: 1) Feature Importance: Identify key descriptors driving target properties. 2) Correlation Matrix: Examine descriptor correlations to determine if properties can be tuned independently. 3) Element Dependence Plot: Evaluate the contribution of each metallic element to guide substitutions.

4. Expected Outcomes

In this section, we will discuss about how to evaluate success based on expected deliverables at each stage.

Stage 1: Systematic DFT Approach for Acid Stability and Activity

- An **automated workflow** integrating existing high-throughput tools (e.g., pymatgen, VASPkit) and custom scripts for input generation, job submission, and post-processing.
- Robust **Pourbaix diagram** construction and visualization capabilities specialized for multi-element oxide systems.

Evaluation Criteria:

- **Throughput-Accuracy Balance:** Ability to efficiently screen a large set of candidate materials while maintaining sufficient accuracy to reliably describe acid stability and OER activity.
- **Automation Reliability:** Minimal manual intervention required, reproducible outputs under consistent computational settings.

Stage 2: Curating an Acid-Stable Surface Slab Dataset

- A **surface slab database** derived from bulk metal oxides identified as acid stable using established methods in Stage 1.
- **Detailed metadata** for each slab (e.g., facet index, termination, and adsorbate configurations).
- **Statistical Overview** summarizing dataset coverage and variability.

Evaluation Criteria:

- **FAIR principles:** The dataset will be findable, accessible, interoperable, and reusable, enabling broad community investigations.

Stage 3: Active Learning Framework with Multicriteria Bayesian Optimization

- An **active learning pipeline** combining multicriteria BO with DFT feedback loops, prioritizing both activity and stability constraints.
- **Experimental validation** (where feasible) or literature evidence to confirm that selected top-ranked candidates align with known high-performance catalysts.

Evaluation Criteria:

- **Adaptive Improvement:** Demonstrable enhancement in predictive accuracy and a smaller search space for viable catalysts after each AL iteration.
- **Efficiency Gains:** Reduced total number of DFT calculations compared to brute-force screening.

Stage 4: Graph-Based Surrogate Model for Structure–Property Predictions

- A graph-based **ML surrogate** capable of predicting OER activity, stability, and other key descriptors from atomic structure.
- **Feature Analysis:** Quantitative insights into how key descriptors (d-band centers, Bader charges, magnetic moments, etc.) influence catalytic performance, as well as the role of elemental composition.

Evaluation Criteria:

- **Predictive Performance:** Low error metrics (e.g., MAE, RMSE).
- **Interpretability**: Clear relationships emerging from SHAP or other model-agnostic methods, guiding atom-level tuning strategies.

5. Reference

(1) Zhang, K.; Zou, R. Advanced Transition Metal-Based OER Electrocatalysts: Current Status, Opportunities, and Challenges. **2021**.

(2) Zhu, Y.; Tang, Z.; Yuan, L.; Li, B.; Shao, Z.; Guo, W. Beyond Conventional Structures: Emerging Complex Metal Oxides for Efficient Oxygen and Hydrogen Electrocatalysis. *Chem. Soc. Rev.* **2024**. https://doi.org/10.1039/D3CS01020A.

(3) Li, Y.; Deng, Y.; Liu, D.; Ji, Q.; Cai, X. Two Dimensional Oxides for Oxygen Evolution Reactions and Related Device Applications. *Mater. Chem. Front.* **2024**, *8* (4), 880–902. https://doi.org/10.1039/D3QM00848G.

(4) Li, L.; Liu, Y.; Chen, Y.; Zhai, W.; Dai, Z. Research Progress on Layered Metal Oxide Electrocatalysts for an Efficient Oxygen Evolution Reaction. *Dalton Trans.* **2024**, *53* (21), 8872–8886. https://doi.org/10.1039/D4DT00619D.

(5) Li, L.; Shao, Q.; Huang, X. Amorphous Oxide Nanostructures for Advanced Electrocatalysis. *Chemistry – A European Journal* **2020**, *26* (18), 3943–3960. https://doi.org/10.1002/chem.201903206.

(6) Cai, H.; Zhang, P.; Li, B.; Zhu, Y.; Zhang, Z.; Guo, W. High-Entropy Oxides for Energy-Related Electrocatalysis. *Materials Today Catalysis* **2024**, *4*, 100039. https://doi.org/10.1016/j.mtcata.2024.100039.

(7) Ding, R.; Liu, J.; Hua, K.; Wang, X.; Zhang, X.; Shao, M.; Chen, J. Leveraging Data Mining, Active Learning, and Domain Adaptation in a Multi-Stage, Machine Learning-Driven Approach for the Efficient Discovery of Advanced Acidic Oxygen Evolution Electrocatalysts.

(8) Jenewein, K. J.; Torresi, L.; Kormányos, A.; Friederich, P.; Cherevko, S. Navigating the Unkown With AI: Multiobjective Bayesian Optimization of Non-Noble Acidic OER Catalysts. **2023**.

(9) Qin, R.; Chen, G.; Feng, X.; Weng, J.; Han, Y. Ru/Ir-Based Electrocatalysts for Oxygen Evolution Reaction in Acidic Conditions: From Mechanisms, Optimizations to Challenges. *Advanced Science* **2024**, *11* (21), 2309364. https://doi.org/10.1002/advs.202309364.

(10) Wan, R.; Yuan, T.; Wang, L.; Li, B.; Liu, M.; Zhao, B. Earth-Abundant Electrocatalysts for Acidic Oxygen Evolution. *Nat Catal* **2024**, *7* (12), 1288–1304. https://doi.org/10.1038/s41929-024-01266-6.

(11) Hong, W. T.; Risch, M.; Stoerzinger, K. A.; Grimaud, A.; Suntivich, J.; Shao-Horn, Y. Toward the Rational Design of Non-Precious Transition Metal Oxides for Oxygen Electrocatalysis. *Energy Environ. Sci.* **2015**, *8* (5), 1404–1427. https://doi.org/10.1039/C4EE03869J.

(12) Wang, Q.; Dastafkan, K.; Zhao, C. Design Strategies for Non-Precious Metal Oxide Electrocatalysts for Oxygen Evolution Reactions. *Current Opinion in Electrochemistry* **2018**, *10*, 16–23. https://doi.org/10.1016/j.coelec.2018.03.015.

(13) Hari Kumar, S. G.; Bozal-Ginesta, C.; Wang, N.; Abed, J.; Shan, C. H.; Yao, Z.; Aspuru-Guzik, A. From Computational Screening to the Synthesis of a Promising OER Catalyst. *Chem. Sci.* **2024**, *15* (27), 10556–10570. https://doi.org/10.1039/D4SC00192C.

(14) Suvarna, M.; Pérez-Ramírez, J. Embracing Data Science in Catalysis Research. *Nat Catal* **2024**, *7* (6), 624–635. https://doi.org/10.1038/s41929-024-01150-3.

(15) Singh, A. K.; Zhou, L.; Shinde, A.; Suram, S. K.; Montoya, J. H.; Winston, D.; Gregoire, J. M.; Persson, K. A. Electrochemical Stability of Metastable Materials. *Chem. Mater.* **2017**, *29* (23), 10159–10167. https://doi.org/10.1021/acs.chemmater.7b03980.

(16) Ong, S. P.; Richards, W. D.; Jain, A.; Hautier, G.; Kocher, M.; Cholia, S.; Gunter, D.; Chevrier, V. L.; Persson, K. A.; Ceder, G. Python Materials Genomics (Pymatgen): A Robust, Open-Source Python Library for Materials Analysis. *Computational Materials Science* **2013**, *68*, 314–319. https://doi.org/10.1016/j.commatsci.2012.10.028.

(17) Fiorentini, V.; Methfessel, M. Extracting Convergent Surface Energies from Slab Calculations. *J. Phys.:*

Condens. Matter **1996**, *8* (36), 6525. https://doi.org/10.1088/0953-8984/8/36/005.
(18) Nørskov, J. K.; Bligaard, T.; Logadottir, A.; Kitchin, J. R.; Chen, J. G.; Pandelov, S.; Stimming, U. Trends in the Exchange Current for Hydrogen Evolution. *J. Electrochem. Soc.* **2005**, *152* (3), J23. https://doi.org/10.1149/1.1856988.
(19) *A Linear Response DFT+U Study of Trends in the Oxygen Evolution Activity of Transition Metal Rutile Dioxides | The Journal of Physical Chemistry C*. https://pubs.acs.org/doi/full/10.1021/jp511426q?src=recsys (accessed 2025-01-09).
(20) Jain, A.; Ong, S. P.; Hautier, G.; Chen, W.; Richards, W. D.; Dacek, S.; Cholia, S.; Gunter, D.; Skinner, D.; Ceder, G.; Persson, K. A. Commentary: The Materials Project: A Materials Genome Approach to Accelerating Materials Innovation. *APL Materials* **2013**, *1* (1), 011002. https://doi.org/10.1063/1.4812323.
(21) Kirklin, S.; Saal, J. E.; Meredig, B.; Thompson, A.; Doak, J. W.; Aykol, M.; Rühl, S.; Wolverton, C. The Open Quantum Materials Database (OQMD): Assessing the Accuracy of DFT Formation Energies. *npj Comput Mater* **2015**, *1* (1), 1–15. https://doi.org/10.1038/npjcompumats.2015.10.
(22) Tran, R.; Lan, J.; Shuaibi, M.; Wood, B. M.; Goyal, S.; Das, A.; Heras-Domingo, J.; Kolluru, A.; Rizvi, A.; Shoghi, N.; Sriram, A.; Therrien, F.; Abed, J.; Voznyy, O.; Sargent, E. H.; Ulissi, Z.; Zitnick, C. L. The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts. *ACS Catal.* **2023**, *13* (5), 3066–3084. https://doi.org/10.1021/acscatal.2c05426.
(23) Abed, J.; Kim, J.; Shuaibi, M.; Wander, B.; Duijf, B.; Mahesh, S.; Lee, H.; Gharakhanyan, V.; Hoogland, S.; Irtem, E.; Lan, J.; Schouten, N.; Vijayakumar, A. U.; Hattrick-Simpers, J.; Kitchin, J. R.; Ulissi, Z. W.; Vugt, A. van; Sargent, E. H.; Sinton, D.; Zitnick, C. L. Open Catalyst Experiments 2024 (OCx24): Bridging Experiments and Computational Models. arXiv November 18, 2024. https://doi.org/10.48550/arXiv.2411.11783.
(24) Back, S.; Tran, K.; Ulissi, Z. W. Discovery of Acid-Stable Oxygen Evolution Catalysts: High-Throughput Computational Screening of Equimolar Bimetallic Oxides. *ACS Appl. Mater. Interfaces* **2020**, *12* (34), 38256–38265. https://doi.org/10.1021/acsami.0c11821.
(25) Gunasooriya, G. T. K. K.; Nørskov, J. K. Analysis of Acid-Stable and Active Oxides for the Oxygen Evolution Reaction. *ACS Energy Lett.* **2020**, *5* (12), 3778–3787. https://doi.org/10.1021/acsenergylett.0c02030.
(26) Wang, Z.; Zheng, Y.-R.; Chorkendorff, I.; Nørskov, J. K. Acid-Stable Oxides for Oxygen Electrocatalysis. *ACS Energy Lett.* **2020**, *5* (9), 2905–2908. https://doi.org/10.1021/acsenergylett.0c01625.
(27) Chen, J.; Ou, P.; Chang, Y.; Zhang, H.; Li, X.-Y.; Sargent, E. H. Adaptive Catalyst Discovery Using Multicriteria Bayesian Optimization with Representation Learning.
(28) Peng, J.; Damewood, J. K.; Karaguesian, J.; Gómez-Bombarelli, R.; Shao-Horn, Y. Navigating Multimetallic Catalyst Space with Bayesian Optimization. *Joule* **2021**, *5* (12), 3069–3071. https://doi.org/10.1016/j.joule.2021.11.011.
(29) Jin, Y.; Kumar, P. V. Bayesian Optimisation for Efficient Material Discovery: A Mini Review. *Nanoscale* **2023**, *15* (26), 10975–10984. https://doi.org/10.1039/D2NR07147A.
(30) Xie, T.; Grossman, J. C. Crystal Graph Convolutional Neural Networks for an Accurate and Interpretable Prediction of Material Properties. arXiv April 6, 2018. https://doi.org/10.48550/arXiv.1710.10324.
(31) Schütt, K. T.; Kindermans, P.-J.; Sauceda, H. E.; Chmiela, S.; Tkatchenko, A.; Müller, K.-R. SchNet: A Continuous-Filter Convolutional Neural Network for Modeling Quantum Interactions. arXiv December 19, 2017. https://doi.org/10.48550/arXiv.1706.08566.
(32) Lundberg, S. M.; Lee, S.-I. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*; Curran Associates, Inc., 2017; Vol. 30.